

## Qualification Pack



# Mould Maker

QP Code: TNSD/CSC/Q0102

Version: 1.0

NSQF Level: 4

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## Qualification Pack

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## Qualification Pack

### TNSD/CSC/Q0102: Mould Maker

#### Brief Job Description

A Mould Maker is responsible for the design, manufacture, and assembly of various moulds and dies, including injection moulds, compression moulds, and die-casting dies. This role requires expertise in advanced machining processes, CAD/CAM software, quality control, and the ability to troubleshoot and repair.

#### Personal Attributes

Upon successful completion, learners will be qualified to become Die Makers for manufacturing dies, moulds, special purpose tools, and die-casting dies. They will gain employment opportunities in engineering, manufacturing, machine tools, automobile, and general machinery industries. This qualification also supports pathways to entrepreneurship.

#### Applicable National Occupational Standards (NOS)

##### Compulsory NOS:

1. [TNSD/CSC/N0109: Perform Non-Conventional Machining and Mould Fabrication Operations](#)
2. [TNSD/CSC/N0110: Design and Draw Multi-day Light and Compression Moulds](#)
3. [TNSD/CSC/N0111: Design Die-Casting Dies and Analyze Defects](#)
4. [TNSD/CSC/N0112: Construct and Troubleshoot Basic Hydraulic and Pneumatic Circuits](#)
5. [TNSD/CSC/N0113: Program and Machine Components using CNC Lathe and Milling Machines](#)
6. [TNSD/CSC/N0114: Perform Basic 2D and 3D Modeling using CREO Software](#)
7. [TNSD/CSC/N0115: Design Complex Assemblies and Moulds using Advanced CREO Features](#)
8. [TNSD/CSC/N0116: Simulate CNC Programs using KELLER Software](#)
9. [TNSD/CSC/N0117: Apply Technical Mathematics for Engineering Calculations](#)
10. [TNSD/CSC/N0118: Understand and Apply Electrical Circuits and Sensors](#)
11. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

#### Qualification Pack (QP) Parameters

## Qualification Pack

|                                                           |                                                                                                 |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                             | Capital Goods                                                                                   |
| <b>Sub-Sector</b>                                         |                                                                                                 |
| <b>Occupation</b>                                         | Bench Fitter, Bench Fitter                                                                      |
| <b>Country</b>                                            | India                                                                                           |
| <b>NSQF Level</b>                                         | 4                                                                                               |
| <b>Credits</b>                                            | 40                                                                                              |
| <b>Aligned to NCO/ISCO/ISIC Code</b>                      | 7222.0500                                                                                       |
| <b>Minimum Educational Qualification &amp; Experience</b> | Previous relevant Qualification of NSQF Level ( Certificate in Tool and Die Bench Fitter – 3.5) |
| <b>Minimum Level of Education for Training in School</b>  |                                                                                                 |
| <b>Pre-Requisite License or Training</b>                  | NA                                                                                              |
| <b>Minimum Job Entry Age</b>                              | 17 Years                                                                                        |
| <b>Last Reviewed On</b>                                   | NA                                                                                              |
| <b>Next Review Date</b>                                   | 07/10/2028                                                                                      |
| <b>NSQC Approval Date</b>                                 | 07/10/2025                                                                                      |
| <b>Version</b>                                            | 1.0                                                                                             |
| <b>Reference code on NQR</b>                              | QG-04-CG-04458-2025-V1-TNSDC                                                                    |
| <b>NQR Version</b>                                        | 1.0                                                                                             |

## Qualification Pack

# TNSD/CSC/N0109: Perform Non-Conventional Machining and Mould Fabrication Operations

## Description

Understand the Non-conventional machining processes used for mould making. Understand the manufacturing of multi-day light injection moulds, compression moulds and simple die-casting dies. Understand the precision matching of mould with die spotting machine.

## Scope

The scope covers the following :

- Understand the non-conventional machining processes
- Understand the manufacturing of Multi-day light injection mould by using CNC and EDM Machine.
- Understand the manufacturing concept of compression mould.

## Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

**PC1.** Understand Non-conventional machining practices for advanced mould making process.

**PC2.** Familiar with CNC Machining for the mould making process.

**PC3.** Manufacture a 2 plate injection mould by using conventional machines, CNC machines and Bench working. Try-out samples.

**PC4.** Manufacture a moulds with ejection system by using conventional machines, CNC machines and Bench working. Try-out samples

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** Understand Non-conventional machining practices for advanced mould making process.

## Generic Skills (GS)

User/individual on the job needs to know how to:

**GS1.** Familiar with CNC Machining for the mould making process.

**GS2.** Manufacture a moulds with ejection system by using conventional machines, CNC machines and Bench working. Try-out samples

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                                                                       | 50           | 200             | 90            | 10         |
| <b>PC1.</b> Understand Non-conventional machining practices for advanced mould making process.                                        | 25           | -               | -             | -          |
| <b>PC2.</b> Familiar with CNC Machining for the mould making process.                                                                 | 25           | -               | -             | -          |
| <b>PC3.</b> Manufacture a 2 plate injection mould by using conventional machines, CNC machines and Bench working. Try-out samples.    | -            | 100             | 90            | 10         |
| <b>PC4.</b> Manufacture a moulds with ejection system by using conventional machines, CNC machines and Bench working. Try-out samples | -            | 100             | -             | -          |
| <b>NOS Total</b>                                                                                                                      | <b>50</b>    | <b>200</b>      | <b>90</b>     | <b>10</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                     |
|----------------------------|---------------------------------------------------------------------|
| <b>NOS Code</b>            | TNSD/CSC/N0109                                                      |
| <b>NOS Name</b>            | Perform Non-Conventional Machining and Mould Fabrication Operations |
| <b>Sector</b>              | Capital Goods                                                       |
| <b>Sub-Sector</b>          |                                                                     |
| <b>Occupation</b>          | Bench Fitter, Bench Fitter                                          |
| <b>NSQF Level</b>          | 4                                                                   |
| <b>Credits</b>             | 19                                                                  |
| <b>Version</b>             | 1.0                                                                 |
| <b>Last Reviewed Date</b>  | 07/10/2025                                                          |
| <b>Next Review Date</b>    | 07/10/2028                                                          |
| <b>NSQC Clearance Date</b> | 07/10/2025                                                          |

## Qualification Pack

# TNSD/CSC/N0110: Design and Draw Multi-day Light and Compression Moulds

## Description

Able to design and draw multi-day light injection moulds, compression mould

## Scope

The scope covers the following :

- Familiarize with the theory and designing of moulds for components with internal, external undercuts and threads.
- Familiarize with concepts of runner-less moulds and hot runner moulds.

## Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the actuation methods of side cores and split cavities
- PC2.** Learn Multi-Day Light mould
- PC3.** Learn Hot runner mould and runner less mould
- PC4.** Understand Injection Moulding defects and remedies
- PC5.** Learn Compression Moulds and transfer moulds
- PC6.** Understand the processes other than injection moulding for thermoplastics.
- PC7.** Learn Surface finishing of mould
- PC8.** understand the Product designing of plastic components
- PC9.** Practice Design and Drawing of moulds with side cores and split cavities
- PC10.** Practice Design and Drawing of multi-day light moulds
- PC11.** Practice Design and Drawing of Compression Moulds
- PC12.** Practice Design and Drawing of Transfer Moulds

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the actuation methods of side cores and split cavities
- KU2.** Learn Hot runner mould and runner less mould
- KU3.** Understand the processes other than injection moulding for thermoplastics.
- KU4.** Understand the Product designing of plastic components

## Generic Skills (GS)

User/individual on the job needs to know how to:

Logo  
Not  
Available



## Qualification Pack

- GS1.** • Learn Hot runner mould and runner less mould  
•  
•
- GS2.** Understand Injection Moulding defects and remedies

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                       | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                        | 100          | 100             | -             | -          |
| <b>PC1.</b> Understand the actuation methods of side cores and split cavities          | 20           | -               | -             | -          |
| <b>PC2.</b> Learn Multi-Day Light mould                                                | 10           | -               | -             | -          |
| <b>PC3.</b> Learn Hot runner mould and runner less mould                               | 5            | -               | -             | -          |
| <b>PC4.</b> Understand Injection Moulding defects and remedies                         | 20           | -               | -             | -          |
| <b>PC5.</b> Learn Compression Moulds and transfer moulds                               | 10           | -               | -             | -          |
| <b>PC6.</b> Understand the processes other than injection moulding for thermoplastics. | 10           | -               | -             | -          |
| <b>PC7.</b> Learn Surface finishing of mould                                           | 10           | -               | -             | -          |
| <b>PC8.</b> understand the Product designing of plastic components                     | 15           | -               | -             | -          |
| <b>PC9.</b> Practice Design and Drawing of moulds with side cores and split cavities   | -            | 30              | -             | -          |
| <b>PC10.</b> Practice Design and Drawing of multi-day light moulds                     | -            | 30              | -             | -          |
| <b>PC11.</b> Practice Design and Drawing of Compression Moulds                         | -            | 20              | -             | -          |
| <b>PC12.</b> Practice Design and Drawing of Transfer Moulds                            | -            | 20              | -             | -          |
| <b>NOS Total</b>                                                                       | <b>100</b>   | <b>100</b>      | <b>-</b>      | <b>-</b>   |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                        |
|----------------------------|--------------------------------------------------------|
| <b>NOS Code</b>            | TNSD/CSC/N0110                                         |
| <b>NOS Name</b>            | Design and Draw Multi-day Light and Compression Moulds |
| <b>Sector</b>              | Capital Goods                                          |
| <b>Sub-Sector</b>          |                                                        |
| <b>Occupation</b>          | Bench Fitter, Bench Fitter                             |
| <b>NSQF Level</b>          | 4                                                      |
| <b>Credits</b>             | 3                                                      |
| <b>Version</b>             | 1.0                                                    |
| <b>Last Reviewed Date</b>  | 07/10/2025                                             |
| <b>Next Review Date</b>    | 07/10/2028                                             |
| <b>NSQC Clearance Date</b> | 07/10/2025                                             |

## Qualification Pack

### TNSD/CSC/N0111: Design Die-Casting Dies and Analyze Defects

#### Description

Able to design and draw simple Die-casting dies. Understand die-casting defects, its causes and remedies.

#### Scope

The scope covers the following :

- Understand the process of die-casting.
- Learn the different calculations pertaining to gate design, force exerted.
- Understand the importance of Die maintenance in die casting.

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the Process description of die casting
- PC2.** Learn effect of metal, machine and die on die casting process
- PC3.** Understand Flow system in die-casting and calculation of force
- PC4.** Familiar with common Die Casting defects and remedies
- PC5.** Practice Design and drawing of different die casting dies.

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the Process description of die casting
- KU2.** Understand Flow system in die-casting and calculation of force

#### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Familiar with common Die Casting defects and remedies

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                           | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                            | 100          | 100             | -             | -          |
| <b>PC1.</b> Understand the Process description of die casting              | 20           | -               | -             | -          |
| <b>PC2.</b> Learn effect of metal, machine and die on die casting process  | 20           | -               | -             | -          |
| <b>PC3.</b> Understand Flow system in die-casting and calculation of force | 30           | -               | -             | -          |
| <b>PC4.</b> Familiar with common Die Casting defects and remedies          | 30           | -               | -             | -          |
| <b>PC5.</b> Practice Design and drawing of different die casting dies.     | -            | 100             | -             | -          |
| <b>NOS Total</b>                                                           | <b>100</b>   | <b>100</b>      | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                             |
|----------------------------|---------------------------------------------|
| <b>NOS Code</b>            | TNSD/CSC/N0111                              |
| <b>NOS Name</b>            | Design Die-Casting Dies and Analyze Defects |
| <b>Sector</b>              | Capital Goods                               |
| <b>Sub-Sector</b>          |                                             |
| <b>Occupation</b>          | Bench Fitter, Bench Fitter                  |
| <b>NSQF Level</b>          | 4                                           |
| <b>Credits</b>             | 3                                           |
| <b>Version</b>             | 1.0                                         |
| <b>Last Reviewed Date</b>  | 07/10/2025                                  |
| <b>Next Review Date</b>    | 07/10/2028                                  |
| <b>NSQC Clearance Date</b> | 07/10/2025                                  |

## Qualification Pack

# TNSD/CSC/N0112: Construct and Troubleshoot Basic Hydraulic and Pneumatic Circuits

## Description

Understand the basics of hydraulics and pneumatics components. Learn to construct a basic circuits and its functioning.

## Scope

The scope covers the following :

- Understand the different types of Pneumatics and Hydraulics components.
- Familiarise with the different symbols used in Pneumatics and Hydraulics circuits.

## Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the different components of Hydraulics and pneumatics system.
- PC2.** List out hydraulic and pneumatic working elements
- PC3.** Design a basic hydraulic and pneumatic circuits
- PC4.** Identify the Hydraulics and Pneumatics components for general and tooling applications
- PC5.** Demonstrate a simple hydraulic circuits & trouble shoot
- PC6.** Demonstrate a simple pneumatic circuits & trouble shoot.

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the different components of Hydraulics and pneumatics system.
- KU2.** Design a basic hydraulic and pneumatic circuits

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Identify the Hydraulics and Pneumatics components for general and tooling applications

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                   | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                                    | 50           | 40              | -             | 10         |
| <b>PC1.</b> Understand the different components of Hydraulics and pneumatics system.               | 10           | -               | -             | -          |
| <b>PC2.</b> List out hydraulic and pneumatic working elements                                      | 15           | -               | -             | -          |
| <b>PC3.</b> Design a basic hydraulic and pneumatic circuits                                        | 15           | -               | -             | -          |
| <b>PC4.</b> Identify the Hydraulics and Pneumatics components for general and tooling applications | 10           | -               | -             | -          |
| <b>PC5.</b> Demonstrate a simple hydraulic circuits & trouble shoot                                | -            | 20              | -             | 5          |
| <b>PC6.</b> Demonstrate a simple pneumatic circuits & trouble shoot.                               | -            | 20              | -             | 5          |
| <b>NOS Total</b>                                                                                   | <b>50</b>    | <b>40</b>       | <b>-</b>      | <b>10</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                   |
|----------------------------|-------------------------------------------------------------------|
| <b>NOS Code</b>            | TNSD/CSC/N0112                                                    |
| <b>NOS Name</b>            | Construct and Troubleshoot Basic Hydraulic and Pneumatic Circuits |
| <b>Sector</b>              | Capital Goods                                                     |
| <b>Sub-Sector</b>          |                                                                   |
| <b>Occupation</b>          | Bench Fitter, Bench Fitter                                        |
| <b>NSQF Level</b>          | 4                                                                 |
| <b>Credits</b>             | 2                                                                 |
| <b>Version</b>             | 1.0                                                               |
| <b>Last Reviewed Date</b>  | 07/10/2025                                                        |
| <b>Next Review Date</b>    | 07/10/2028                                                        |
| <b>NSQC Clearance Date</b> | 07/10/2025                                                        |

## Qualification Pack

# TNSD/CSC/N0113: Program and Machine Components using CNC Lathe and Milling Machines

## Description

Able to do the programming on CNC lathe, Milling machine. Able to do the machining of components on CNC lathe, Milling machine.

## Scope

The scope covers the following :

- Understanding the concepts and CNC systems
- Familiar with G-codes and M-codes used in programming
- Learn the Selection of right feed and speed for different cutting tool materials and usage of right inserts and tool holders.

## Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

**PC1.** understand the basic concepts and constructional features of CNC machines

**PC2.** Understand the coordinate system & codes

**PC3.** Learn the concept of offset and different modes of operation

**PC4.** Understand the different types of tool holders and inserts used in CNC machines.

**PC5.** Program and do machining a part on CNC lathe.

**PC6.** Program and do machining a part on CNC Milling.

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** understand the basic concepts and constructional features of CNC machines

**KU2.** Understand the coordinate system & codes

## Generic Skills (GS)

User/individual on the job needs to know how to:

**GS1.** Program and do machining a part on CNC lathe.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                             | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                              | 100          | 90              | -             | 10         |
| <b>PC1.</b> understand the basic concepts and constructional features of CNC machines        | 15           | -               | -             | -          |
| <b>PC2.</b> Understand the coordinate system & codes                                         | 15           | -               | -             | -          |
| <b>PC3.</b> Learn the concept of offset and different modes of operation                     | 15           | -               | -             | -          |
| <b>PC4.</b> Understand the different types of tool holders and inserts used in CNC machines. | 15           | -               | -             | -          |
| <b>PC5.</b> Program and do machining a part on CNC lathe.                                    | 20           | 45              | -             | 5          |
| <b>PC6.</b> Program and do machining a part on CNC Milling.                                  | 20           | 45              | -             | 5          |
| <b>NOS Total</b>                                                                             | <b>100</b>   | <b>90</b>       | <b>-</b>      | <b>10</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                     |
|----------------------------|---------------------------------------------------------------------|
| <b>NOS Code</b>            | TNSD/CSC/N0113                                                      |
| <b>NOS Name</b>            | Program and Machine Components using CNC Lathe and Milling Machines |
| <b>Sector</b>              | Capital Goods                                                       |
| <b>Sub-Sector</b>          |                                                                     |
| <b>Occupation</b>          | Bench Fitter, Bench Fitter                                          |
| <b>NSQF Level</b>          | 4                                                                   |
| <b>Credits</b>             | 3                                                                   |
| <b>Version</b>             | 1.0                                                                 |
| <b>Last Reviewed Date</b>  | 07/10/2025                                                          |
| <b>Next Review Date</b>    | 07/10/2028                                                          |
| <b>NSQC Clearance Date</b> | 07/10/2025                                                          |

## Qualification Pack

### TNSD/CSC/N0114: Perform Basic 2D and 3D Modeling using CREO Software

#### Description

Able to understand the basics for using Creo software Able to do 2D and 3D modelling.

#### Scope

The scope covers the following :

- Familiar with different toolbars and functions of CREO software.
- Understanding Basics of 2D Drawing, Modelling (Surface &Solid) by Creo software.
- Understanding the tools used for Surface modelling

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

**PC1.** Practice using Creo tools for sketching

**PC2.** Practice using Creo tools for Basic Modelling

**PC3.** Identify and practice the method of advanced modelling

**PC4.** Perform surface modelling using creo software.

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** Practice using Creo tools for sketching

**KU2.** Practice using Creo tools for Basic Modelling

#### Generic Skills (GS)

User/individual on the job needs to know how to:

**GS1.** Identify and practice the method of advanced modelling

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                   | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                    | -            | 90              | -             | 10         |
| <b>PC1.</b> Practice using Creo tools for sketching                | -            | 30              | -             | 5          |
| <b>PC2.</b> Practice using Creo tools for Basic Modelling          | -            | 20              | -             | -          |
| <b>PC3.</b> Identify and practice the method of advanced modelling | -            | 20              | -             | 5          |
| <b>PC4.</b> Perform surface modelling using creo software.         | -            | 20              | -             | -          |
| <b>NOS Total</b>                                                   | -            | 90              | -             | 10         |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                      |
|----------------------------|------------------------------------------------------|
| <b>NOS Code</b>            | TNSD/CSC/N0114                                       |
| <b>NOS Name</b>            | Perform Basic 2D and 3D Modeling using CREO Software |
| <b>Sector</b>              | Capital Goods                                        |
| <b>Sub-Sector</b>          |                                                      |
| <b>Occupation</b>          | Bench Fitter, Bench Fitter                           |
| <b>NSQF Level</b>          | 4                                                    |
| <b>Credits</b>             | 2                                                    |
| <b>Version</b>             | 1.0                                                  |
| <b>Last Reviewed Date</b>  | 07/10/2025                                           |
| <b>Next Review Date</b>    | 07/10/2028                                           |
| <b>NSQC Clearance Date</b> | 07/10/2025                                           |

## Qualification Pack

### TNSD/CSC/N0115: Design Complex Assemblies and Moulds using Advanced CREO Features

#### Description

Draw different assemblies and Surface modelling using CERO software. Designing and drafting of moulds using CERO software.

#### Scope

The scope covers the following :

- Learn different functions of CREO software for drawing different assemblies.
- Understand the steps for core and cavity Extraction.
- Familiarize CREO software for drawing of moulds.

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

**PC1.** Perform Concept of assembling and detailed drawing using creo software

**PC2.** Demonstrate Drafting with GD&T using creo software.

**PC3.** Demonstrate Extraction of core and cavity using creo software.

**PC4.** Perform Design of standard mould base using creo software.

**PC5.** Perform Design and drawing of mould using creo software.

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** Perform Concept of assembling and detailed drawing using creo software.

**KU2.** Demonstrate Drafting with GD&T using creo software.

#### Generic Skills (GS)

User/individual on the job needs to know how to:

**GS1.** Perform Design of standard mould base using creo software.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                   | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                    | -            | 90              | -             | 10         |
| <b>PC1.</b> Perform Concept of assembling and detailed drawing using creo software | -            | 10              | -             | -          |
| <b>PC2.</b> Demonstrate Drafting with GD&T using creo software.                    | -            | 10              | -             | 5          |
| <b>PC3.</b> Demonstrate Extraction of core and cavity using creo software.         | -            | 30              | -             | 5          |
| <b>PC4.</b> Perform Design of standard mould base using creo software.             | -            | 10              | -             | -          |
| <b>PC5.</b> Perform Design and drawing of mould using creo software.               | -            | 30              | -             | -          |
| <b>NOS Total</b>                                                                   | -            | 90              | -             | 10         |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                   |
|----------------------------|-------------------------------------------------------------------|
| <b>NOS Code</b>            | TNSD/CSC/N0115                                                    |
| <b>NOS Name</b>            | Design Complex Assemblies and Moulds using Advanced CREO Features |
| <b>Sector</b>              | Capital Goods                                                     |
| <b>Sub-Sector</b>          |                                                                   |
| <b>Occupation</b>          | Bench Fitter, Bench Fitter                                        |
| <b>NSQF Level</b>          | 4                                                                 |
| <b>Credits</b>             | 2                                                                 |
| <b>Version</b>             | 1.0                                                               |
| <b>Last Reviewed Date</b>  | 07/10/2025                                                        |
| <b>Next Review Date</b>    | 07/10/2028                                                        |
| <b>NSQC Clearance Date</b> | 07/10/2025                                                        |

## Qualification Pack

### TNSD/CSC/N0116: Simulate CNC Programs using KELLER Software

#### Description

Simulation of CNC program by Keller software.

#### Scope

The scope covers the following :

- Learn different tools and functions of CNC Machines using KELLER software.
- Learn to program and do simulation by using KELLER software.

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Practice working environment of CNC machines and identify the parts using keller software.
- PC2.** Do basic programming with right Feed & Speed and understand the Cycle time on CNC lathe using keller software.
- PC3.** Do basic programming with right Feed & Speed and understand the Cycle time on CNC Milling using keller software.
- PC4.** Demonstrate Various ISO cycles and its implementation in CNC Lathe.
- PC5.** Demonstrate the Simulation of Converting raw material to finished part on CNC Milling, Lathe using keller software.

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Practice working environment of CNC machines and identify the parts using keller software.
- KU2.** Do basic programming with right Feed & Speed and understand the Cycle time on CNC lathe using keller software.

#### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Demonstrate the Simulation of Converting raw material to finished part on CNC Milling, Lathe using keller software.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                                                                 | -            | 90              | -             | 10         |
| <b>PC1.</b> Practice working environment of CNC machines and identify the parts using keller software.                          | -            | 10              | -             | -          |
| <b>PC2.</b> Do basic programming with right Feed & Speed and understand the Cycle time on CNC lathe using keller software.      | -            | 20              | -             | 5          |
| <b>PC3.</b> Do basic programming with right Feed & Speed and understand the Cycle time on CNC Milling using keller software.    | -            | 20              | -             | -          |
| <b>PC4.</b> Demonstrate Various ISO cycles and its implementation in CNC Lathe.                                                 | -            | 20              | -             | 5          |
| <b>PC5.</b> Demonstrate the Simulation of Converting raw material to finished part on CNC Milling, Lathe using keller software. | -            | 20              | -             | -          |
| <b>NOS Total</b>                                                                                                                | -            | 90              | -             | 10         |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                             |
|----------------------------|---------------------------------------------|
| <b>NOS Code</b>            | TNSD/CSC/N0116                              |
| <b>NOS Name</b>            | Simulate CNC Programs using KELLER Software |
| <b>Sector</b>              | Capital Goods                               |
| <b>Sub-Sector</b>          |                                             |
| <b>Occupation</b>          | Bench Fitter, Bench Fitter                  |
| <b>NSQF Level</b>          | 4                                           |
| <b>Credits</b>             | 2                                           |
| <b>Version</b>             | 1.0                                         |
| <b>Last Reviewed Date</b>  | 07/10/2025                                  |
| <b>Next Review Date</b>    | 07/10/2028                                  |
| <b>NSQC Clearance Date</b> | 07/10/2025                                  |

## Qualification Pack

# TNSD/CSC/N0117: Apply Technical Mathematics for Engineering Calculations

## Description

Understand the concept of basic mathematics used for doing calculations related to engineering application and general purpose.

## Scope

The scope covers the following :

- Understand the applications of Trigonometrical functions in engineering field.
- Understand the basic concept & application of calculus.
- Understand the concept of Analytical Geometry.

## Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Learn the Concepts of fundamental Analytical geometry
- PC2.** Find the slope of line , parallel and perpendicular lines
- PC3.** Find the distance between the vertices Transversal of line
- PC4.** Calculate the area of triangles and quadrilaterals , mid points, Equation of lines and related problems
- PC5.** Calculate the Centroid , in center, Ortho center of a triangles and related problems.
- PC6.** Understand the basic concept & application of calculus
- PC7.** Solve the linear and non linear equations , Ordinary differential & partial differential equations.
- PC8.** Find the order and degree of the differential equations, solution of ODE& PDE
- PC9.** Learn the Differential rule , Chain rule, Bernoulli rule and related problems
- PC10.** Learn Basic concept of principle of Trigonometry and its applications
- PC11.** Solve the problems using trigonometry application
- PC12.** Find the Solution of acute , obtuse angled triangles and related problems
- PC13.** Find the values of angles by using trigonometric table values, Angle value for the special angles, Solution of circumscribed triangles by sine rule and practice problems

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Learn the Concepts of fundamental Analytical geometry
- KU2.** Calculate the area of triangles and quadrilaterals , mid points, Equation of lines and related problems.

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Available



## Qualification Pack

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Solve the linear and non linear equations , Ordinary differential & partial differential equations.
- GS2.** Learn the Differential rule ,Chain rule, Bernoulli rule and related problems

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                       | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                                                                                                                        | 100          | -               | -             | -          |
| <b>PC1.</b> Learn the Concepts of fundamental Analytical geometry                                                                                                                      | 5            | -               | -             | -          |
| <b>PC2.</b> Find the slope of line , parallel and perpendicular lines                                                                                                                  | 5            | -               | -             | -          |
| <b>PC3.</b> Find the distance between the vertices Transversal of line                                                                                                                 | 5            | -               | -             | -          |
| <b>PC4.</b> Calculate the area of triangles and quadrilaterals , mid points, Equation of lines and related problems                                                                    | 10           | -               | -             | -          |
| <b>PC5.</b> Calculate the Centroid , in center, Ortho center of a triangles and related problems.                                                                                      | 10           | -               | -             | -          |
| <b>PC6.</b> Understand the basic concept & application of calculus                                                                                                                     | 5            | -               | -             | -          |
| <b>PC7.</b> Solve the linear and non linear equations , Ordinary differential & partial differential equations.                                                                        | 10           | -               | -             | -          |
| <b>PC8.</b> Find the order and degree of the differential equations, solution of ODE& PDE                                                                                              | 10           | -               | -             | -          |
| <b>PC9.</b> Learn the Differential rule , Chain rule, Bernoulli rule and related problems                                                                                              | 10           | -               | -             | -          |
| <b>PC10.</b> Learn Basic concept of principle of Trigonometry and its applications                                                                                                     | 5            | -               | -             | -          |
| <b>PC11.</b> Solve the problems using trigonometry application                                                                                                                         | 10           | -               | -             | -          |
| <b>PC12.</b> Find the Solution of acute , obtuse angled triangles and related problems                                                                                                 | 10           | -               | -             | -          |
| <b>PC13.</b> Find the values of angles by using trigonometric table values, Angle value for the special angles, Solution of circumscribed triangles by sine rule and practice problems | 5            | -               | -             | -          |

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Not  
Available



### Qualification Pack

| Assessment Criteria for Outcomes | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------|--------------|-----------------|---------------|------------|
| NOS Total                        | 100          | -               | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                          |
|----------------------------|----------------------------------------------------------|
| <b>NOS Code</b>            | TNSD/CSC/N0117                                           |
| <b>NOS Name</b>            | Apply Technical Mathematics for Engineering Calculations |
| <b>Sector</b>              | Capital Goods                                            |
| <b>Sub-Sector</b>          |                                                          |
| <b>Occupation</b>          | Bench Fitter, Bench Fitter                               |
| <b>NSQF Level</b>          | 4                                                        |
| <b>Credits</b>             | 1                                                        |
| <b>Version</b>             | 1.0                                                      |
| <b>Last Reviewed Date</b>  | 07/10/2025                                               |
| <b>Next Review Date</b>    | 07/10/2028                                               |
| <b>NSQC Clearance Date</b> | 07/10/2025                                               |

## Qualification Pack

### TNSD/CSC/N0118: Understand and Apply Electrical Circuits and Sensors

#### Description

Understanding the applications of basic electrical circuits Understanding the applications of sensors.

#### Scope

The scope covers the following :

- Understand the construction of electric circuits.
- Understand the concepts of sensors

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the Basic of Electricity - Voltage -Current
- PC2.** Learn Electromagnetic Induction Principle-Working of electrical components like Solenoids, Motors, Generator, Switches, Fuse and Circuit Breakers
- PC3.** Learn the Fundamentals of Sensor and its types
- PC4.** Understand the Application of sensors
- PC5.** Understand the basic Electricals and Electronics components for Tooling applications

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the Basic of Electricity Voltage Current
- KU2.** Understand the Application of sensors

#### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Understand the basic Electricals and Electronics components for Tooling applications

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                              | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                                                                                               | 50           | -               | -             | -          |
| <b>PC1.</b> Understand the Basic of Electricity - Voltage -Current                                                                                            | 10           | -               | -             | -          |
| <b>PC2.</b> Learn Electromagnetic Induction Principle-Working of electrical components like Solenoids, Motors, Generator, Switches, Fuse and Circuit Breakers | 10           | -               | -             | -          |
| <b>PC3.</b> Learn the Fundamentals of Sensor and its types                                                                                                    | 10           | -               | -             | -          |
| <b>PC4.</b> Understand the Application of sensors                                                                                                             | 10           | -               | -             | -          |
| <b>PC5.</b> Understand the basic Electricals and Electronics components for Tooling applications                                                              | 10           | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                              | <b>50</b>    | -               | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                      |
|----------------------------|------------------------------------------------------|
| <b>NOS Code</b>            | TNSD/CSC/N0118                                       |
| <b>NOS Name</b>            | Understand and Apply Electrical Circuits and Sensors |
| <b>Sector</b>              | Capital Goods                                        |
| <b>Sub-Sector</b>          |                                                      |
| <b>Occupation</b>          | Bench Fitter, Bench Fitter                           |
| <b>NSQF Level</b>          | 4                                                    |
| <b>Credits</b>             | 1                                                    |
| <b>Version</b>             | 1.0                                                  |
| <b>Last Reviewed Date</b>  | 07/10/2025                                           |
| <b>Next Review Date</b>    | 07/10/2028                                           |
| <b>NSQC Clearance Date</b> | 07/10/2025                                           |

## Qualification Pack

### DGT/VSQ/N0102: Employability Skills (60 Hours)

#### Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

#### Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

#### Elements and Performance Criteria

##### *Introduction to Employability Skills*

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

##### *Constitutional values – Citizenship*

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

##### *Becoming a Professional in the 21st Century*

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

##### *Basic English Skills*

To be competent, the user/individual on the job must be able to:

## Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

### *Career Development & Goal Setting*

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

### *Communication Skills*

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

### *Diversity & Inclusion*

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

### *Financial and Legal Literacy*

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

### *Essential Digital Skills*

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

### *Entrepreneurship*

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

### *Customer Service*

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

## Qualification Pack

**PC28.** follow appropriate hygiene and grooming standards

*Getting ready for apprenticeship & Jobs*

To be competent, the user/individual on the job must be able to:

**PC29.** create a professional Curriculum vitae (Résumé)

**PC30.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

**PC31.** apply to identified job openings using offline /online methods as per requirement

**PC32.** answer questions politely, with clarity and confidence, during recruitment and selection

**PC33.** identify apprenticeship opportunities and register for it as per guidelines and requirements

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** need for employability skills and different learning and employability related portals

**KU2.** various constitutional and personal values

**KU3.** different environmentally sustainable practices and their importance

**KU4.** Twenty first (21st) century skills and their importance

**KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

**KU6.** importance of career development and setting long- and short-term goals

**KU7.** about effective communication

**KU8.** POSH Act

**KU9.** Gender sensitivity and inclusivity

**KU10.** different types of financial institutes, products, and services

**KU11.** how to compute income and expenditure

**KU12.** importance of maintaining safety and security in offline and online financial transactions

**KU13.** different legal rights and laws

**KU14.** different types of digital devices and the procedure to operate them safely and securely

**KU15.** how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

**KU16.** how to identify business opportunities

**KU17.** types and needs of customers

**KU18.** how to apply for a job and prepare for an interview

**KU19.** apprenticeship scheme and the process of registering on apprenticeship portal

## Generic Skills (GS)

User/individual on the job needs to know how to:

**GS1.** read and write different types of documents/instructions/correspondence

**GS2.** communicate effectively using appropriate language in formal and informal settings

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## Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Introduction to Employability Skills</i>                                                                                                                                                                                                                                                                      | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC1.</b> identify employability skills required for jobs in various industries                                                                                                                                                                                                                                | -            | -               | -             | -          |
| <b>PC2.</b> identify and explore learning and employability portals                                                                                                                                                                                                                                              | -            | -               | -             | -          |
| <i>Constitutional values – Citizenship</i>                                                                                                                                                                                                                                                                       | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC3.</b> recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.                                                               | -            | -               | -             | -          |
| <b>PC4.</b> follow environmentally sustainable practices                                                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <i>Becoming a Professional in the 21st Century</i>                                                                                                                                                                                                                                                               | <b>2</b>     | <b>4</b>        | -             | -          |
| <b>PC5.</b> recognize the significance of 21st Century Skills for employment                                                                                                                                                                                                                                     | -            | -               | -             | -          |
| <b>PC6.</b> practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life | -            | -               | -             | -          |
| <i>Basic English Skills</i>                                                                                                                                                                                                                                                                                      | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC7.</b> use basic English for everyday conversation in different contexts, in person and over the telephone                                                                                                                                                                                                  | -            | -               | -             | -          |
| <b>PC8.</b> read and understand routine information, notes, instructions, mails, letters etc. written in English                                                                                                                                                                                                 | -            | -               | -             | -          |
| <b>PC9.</b> write short messages, notes, letters, e-mails etc. in English                                                                                                                                                                                                                                        | -            | -               | -             | -          |
| <i>Career Development &amp; Goal Setting</i>                                                                                                                                                                                                                                                                     | <b>1</b>     | <b>2</b>        | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> understand the difference between job and career                                                         | -            | -               | -             | -          |
| <b>PC11.</b> prepare a career development plan with short- and long-term goals, based on aptitude                     | -            | -               | -             | -          |
| <i>Communication Skills</i>                                                                                           | <b>2</b>     | <b>2</b>        | -             | -          |
| <b>PC12.</b> follow verbal and non-verbal communication etiquette and active listening techniques in various settings | -            | -               | -             | -          |
| <b>PC13.</b> work collaboratively with others in a team                                                               | -            | -               | -             | -          |
| <i>Diversity &amp; Inclusion</i>                                                                                      | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC14.</b> communicate and behave appropriately with all genders and PwD                                            | -            | -               | -             | -          |
| <b>PC15.</b> escalate any issues related to sexual harassment at workplace according to POSH Act                      | -            | -               | -             | -          |
| <i>Financial and Legal Literacy</i>                                                                                   | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC16.</b> select financial institutions, products and services as per requirement                                  | -            | -               | -             | -          |
| <b>PC17.</b> carry out offline and online financial transactions, safely and securely                                 | -            | -               | -             | -          |
| <b>PC18.</b> identify common components of salary and compute income, expenses, taxes, investments etc                | -            | -               | -             | -          |
| <b>PC19.</b> identify relevant rights and laws and use legal aids to fight against legal exploitation                 | -            | -               | -             | -          |
| <i>Essential Digital Skills</i>                                                                                       | <b>3</b>     | <b>4</b>        | -             | -          |
| <b>PC20.</b> operate digital devices and carry out basic internet operations securely and safely                      | -            | -               | -             | -          |
| <b>PC21.</b> use e- mail and social media platforms and virtual collaboration tools to work effectively               | -            | -               | -             | -          |
| <b>PC22.</b> use basic features of word processor, spreadsheets, and presentations                                    | -            | -               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Entrepreneurship</i>                                                                                                                                                          | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC23.</b> identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research                                       | -            | -               | -             | -          |
| <b>PC24.</b> develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion                                                      | -            | -               | -             | -          |
| <b>PC25.</b> identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity                                           | -            | -               | -             | -          |
| <i>Customer Service</i>                                                                                                                                                          | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC26.</b> identify different types of customers                                                                                                                               | -            | -               | -             | -          |
| <b>PC27.</b> identify and respond to customer requests and needs in a professional manner.                                                                                       | -            | -               | -             | -          |
| <b>PC28.</b> follow appropriate hygiene and grooming standards                                                                                                                   | -            | -               | -             | -          |
| <i>Getting ready for apprenticeship &amp; Jobs</i>                                                                                                                               | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC29.</b> create a professional Curriculum vitae (Résumé)                                                                                                                     | -            | -               | -             | -          |
| <b>PC30.</b> search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively | -            | -               | -             | -          |
| <b>PC31.</b> apply to identified job openings using offline /online methods as per requirement                                                                                   | -            | -               | -             | -          |
| <b>PC32.</b> answer questions politely, with clarity and confidence, during recruitment and selection                                                                            | -            | -               | -             | -          |
| <b>PC33.</b> identify apprenticeship opportunities and register for it as per guidelines and requirements                                                                        | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                 | <b>20</b>    | <b>30</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                 |
|----------------------------|---------------------------------|
| <b>NOS Code</b>            | DGT/VSQ/N0102                   |
| <b>NOS Name</b>            | Employability Skills (60 Hours) |
| <b>Sector</b>              | Cross Sectoral                  |
| <b>Sub-Sector</b>          | Professional Skills             |
| <b>Occupation</b>          | Employability                   |
| <b>NSQF Level</b>          | 4                               |
| <b>Credits</b>             | 2                               |
| <b>Version</b>             | 1.0                             |
| <b>Last Reviewed Date</b>  | 18/02/2025                      |
| <b>Next Review Date</b>    | 18/02/2028                      |
| <b>NSQC Clearance Date</b> | 18/02/2025                      |

## Assessment Guidelines and Assessment Weightage

### Assessment Guidelines

#### 1. Assessment System Overview:

- Batches assigned to the assessment agencies for conducting the assessment on SIP or email
- Assessment agencies send the assessment confirmation to VTP/TC looping TNSDC
- Assessment agency deploys the ToA certified Assessor for executing the assessment
- TNSDC monitors the assessment process & records

#### 2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 2 Assessors.
- Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.

#### 3. Assessment Quality Assurance levels/Framework:

## Qualification Pack

- Question bank is created by the Subject Matter Experts (SME) are verified by the other SME
- Questions are mapped to the specified assessment criteria
- Assessor must be ToA certified & trainer must be ToT Certified

### 4. Types of evidence or evidence-gathering protocol:

- Time-stamped & geotagged reporting of the assessor from assessment location
- Centre photographs with signboards and scheme specific branding

### 5. Method of verification or validation:

- Surprise visit to the assessment location

### 6. Method for assessment documentation, archiving, and access

- Hard copies of the documents are stored

### On the Job:

1. Each module will be assessed separately.
2. The candidate must score 60% in each module to successfully complete the OJT.
3. Tools of Assessment that will be used for assessing whether the candidate is having desired skills and etiquette of dealing with customers, understanding needs & requirements, assessing the customer and perform Soft Skills effectively:
  - Videos of Trainees during OJT
4. Assessment of each Module will ensure that the candidate is able to:
  - Effective engagement with the customers
  - Understand the working of various tools and equipment

## Minimum Aggregate Passing % at QP Level : 60

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

## Assessment Weightage

## Qualification Pack

### Compulsory NOS

| National Occupational Standards                                                    | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage  |
|------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|------------|
| TNSD/CSC/N0109.Perform Non-Conventional Machining and Mould Fabrication Operations | 50           | 200             | 90            | 10         | 350         | 15         |
| TNSD/CSC/N0110.Design and Draw Multi-day Light and Compression Moulds              | 100          | 100             | 0             | 0          | 200         | 10         |
| TNSD/CSC/N0111.Design Die-Casting Dies and Analyze Defects                         | 100          | 100             | 0             | 0          | 200         | 10         |
| TNSD/CSC/N0112.Construct and Troubleshoot Basic Hydraulic and Pneumatic Circuits   | 50           | 40              | 0             | 10         | 100         | 10         |
| TNSD/CSC/N0113.Program and Machine Components using CNC Lathe and Milling Machines | 100          | 90              | 0             | 10         | 200         | 10         |
| TNSD/CSC/N0114.Perform Basic 2D and 3D Modeling using CREO Software                | 0            | 90              | 0             | 10         | 100         | 10         |
| TNSD/CSC/N0115.Design Complex Assemblies and Moulds using Advanced CREO Features   | 0            | 90              | 0             | 10         | 100         | 10         |
| TNSD/CSC/N0116.Simulate CNC Programs using KELLER Software                         | 0            | 90              | 0             | 10         | 100         | 10         |
| TNSD/CSC/N0117.Apply Technical Mathematics for Engineering Calculations            | 100          | 0               | 0             | 0          | 100         | 5          |
| TNSD/CSC/N0118.Understand and Apply Electrical Circuits and Sensors                | 50           | 0               | 0             | 0          | 50          | 5          |
| DGT/VSQ/N0102.Employability Skills (60 Hours)                                      | 20           | 30              | -             | -          | 50          | 5          |
| <b>Total</b>                                                                       | <b>570</b>   | <b>830</b>      | <b>90</b>     | <b>60</b>  | <b>1550</b> | <b>100</b> |

Logo  
Not  
Available



## Qualification Pack

### Acronyms

|             |                                                 |
|-------------|-------------------------------------------------|
| <b>NOS</b>  | National Occupational Standard(s)               |
| <b>NSQF</b> | National Skills Qualifications Framework        |
| <b>QP</b>   | Qualifications Pack                             |
| <b>TVET</b> | Technical and Vocational Education and Training |

## Qualification Pack

### Glossary

|                                              |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                      |
| <b>Sub-sector</b>                            | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                     |
| <b>Occupation</b>                            | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                |
| <b>Job role</b>                              | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                |
| <b>Occupational Standards (OS)</b>           | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts. |
| <b>Performance Criteria (PC)</b>             | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                    |
| <b>National Occupational Standards (NOS)</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                       |
| <b>Qualifications Pack (QP)</b>              | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                       |
| <b>Unit Code</b>                             | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                        |
| <b>Unit Title</b>                            | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                        |
| <b>Description</b>                           | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                   |
| <b>Scope</b>                                 | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                            |

## Qualification Pack

|                                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge and Understanding (KU)</b> | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                                                                                               |
| <b>Organisational Context</b>           | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>              | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills (GS)</b> | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                        | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                          | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |